

SAT Math Mastery · Week 1B**Lines, Slope & Linear Functions**

Time	120 minutes: 90 min solving + 30 min self-check and error log
Topics	slope from points/graphs/tables · forms of a line · intercepts · parallel & perpendicular · translations · meaning of slope/intercept · linear function notation
Name / Date	_____

1 · SKILLS & IDEAS

- S1.** Find slope from two points, a graph, a table, a standard-form equation, or a verbal rate.
- S2.** Write and convert between slope-intercept, point-slope, standard and intercept forms.
- S3.** Find x - and y -intercepts and use them (areas, conditions, context meaning).
- S4.** Parallel and perpendicular lines, including directly from $Ax + By = C$.
- S5.** Translate a line (up/down/left/right) and track slope and intercepts.
- S6.** Interpret slope and intercept in context; build $f(x) = mx + b$ from data and solve for inputs.
- S7.** Coordinate reasoning: collinearity, midpoint, perpendicular bisector, triangles cut by axes.
- S8.** Parameter conditions on lines (a constant fixes slope, intercept, perpendicularity or area).

2 · REQUIRED FACTS

- Slope = $\frac{\text{rise}}{\text{run}}$ = constant rate of change; the y -intercept is the value at $x = 0$; the x -intercept is where $y = 0$.
- Horizontal line $y = c$: slope 0. Vertical line $x = c$: slope undefined.
- Parallel $\iff$ equal slopes (different intercepts). Perpendicular $\iff$ slopes are negative reciprocals (product -1).
- A function is linear if equal steps in x give equal steps in y (check every gap in a table).
- Shifting a graph up k adds k to y ; slope never changes under a translation.
- Two points determine a line; on a graph read only points that sit exactly on grid intersections.

3 · REQUIRED FORMULAS

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y = mx + b, \quad y - y_1 = m(x - x_1), \quad \frac{x}{a} + \frac{y}{b} = 1$$

$$Ax + By = C: \quad m = -\frac{A}{B}, \quad x\text{-int } \frac{C}{A}, \quad y\text{-int } \frac{C}{B}, \quad \text{area with axes} = \frac{1}{2} \left| \frac{C}{A} \cdot \frac{C}{B} \right|$$

$$\parallel: A_1B_2 = A_2B_1, \quad \perp: A_1A_2 + B_1B_2 = 0, \quad \text{midpoint} = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right), \quad f(a + d) - f(a) = md$$

4 · MINI THEORY SHEET

Choose the form by the data.

Given	Fastest route
two points	$m = \frac{\Delta y}{\Delta x}$, then point-slope with either point
slope and y -intercept	$y = mx + b$ directly
both intercepts $(a, 0), (0, b)$	$\frac{x}{a} + \frac{y}{b} = 1$, or $m = -\frac{b}{a}$
$Ax + By = C$	read $m = -\frac{A}{B}$; set $x = 0$ or $y = 0$ for intercepts
a table	m from any two rows; confirm the rest have the same m

Perpendicular in standard form. Swap the coefficients and negate one: $\perp$ to $Ax + By = C$ is $Bx - Ay = C'$; find C' from the given point.

Parallel in standard form. Keep A, B (or scale both), change only C .

Translation. Down k : $b \rightarrow b - k$, and the x -intercept moves by $\frac{k}{m}$. Same slope, always.

Meaning. In $y = mx + b$: m = change in y per 1 unit of x ; b = starting value. Units of m are (units of y)/(units of x).

Constant conditions. Turn the condition (slope, perpendicular, area, collinear) into one equation in the constant, then solve.

Whole-number contexts. Solve $f(x) = \text{target}$; if x must be a whole day/trip/session, round to the first value that *reaches* the target.

5 · GUIDED EXAMPLES

Example 1 — Two points $\rightarrow$ equation $\rightarrow$ intercepts

A line passes through $(-1, 7)$ and $(3, -1)$. Find its equation and both intercepts.

$$m = \frac{-1 - 7}{3 - (-1)} = \frac{-8}{4} = -2. \text{ Point-slope: } y - 7 = -2(x + 1) \Rightarrow y = -2x + 5.$$

$$y\text{-intercept } 5; x\text{-intercept: } 0 = -2x + 5 \Rightarrow x = \frac{5}{2}.$$

Example 2 — Perpendicular through a point (standard form)

Line ℓ : $4x - 3y = 12$. Write the line through $(2, -1)$ perpendicular to ℓ .

Swap and negate: $3x + 4y = C'$. Plug in $(2, -1)$: $C' = 3(2) + 4(-1) = 2$. Answer: $3x + 4y = 2$.

Example 3 — Meaning of slope and intercept

A gym plan costs \$340 after 2 months and \$640 after 5 months (linear). Write $C(m)$ and interpret its parts.

$$\text{slope} = \frac{640 - 340}{5 - 2} = 100. \quad C(m) = 100m + b; \quad 340 = 200 + b \Rightarrow b = 140. \quad \text{So } C(m) = 100m + 140.$$

100 = dollars per month; 140 = joining fee (cost at $m = 0$).

6 · PRACTICE SET — 25 PROBLEMS · EASY → MEDIUM → HARD → SAT CHALLENGE

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal).

1. EASY · NO CALCULATOR NEEDED

The function f is defined by $f(x) = 15x + 10$. What is the value of $f(x)$ when $x = 2$?

- (A) 30 (B) 40
(C) 50 (D) 27

2. EASY · NO CALCULATOR NEEDED

In the xy -plane, the graph of $y = f(x)$ is a line that passes through $(0, 21)$ and has a slope of 4. Which equation defines f ?

- (A) $f(x) = -4x + 21$ (B) $f(x) = 21x + 4$
(C) $f(x) = 4x + 21$ (D) $f(x) = 4x - 21$

3. EASY · NO CALCULATOR NEEDED

A line has a slope of -3 and passes through the point $(5, -2)$. What is the y -intercept of the line?

4. EASY · NO CALCULATOR NEEDED

What is the x -intercept of the line given by $3x + 4y = 24$?

- (A) $(8, 0)$ (B) $(0, 6)$
(C) $(0, 8)$ (D) $(6, 0)$

5.

EASY · NO CALCULATOR NEEDED

The function g is defined by $g(x) = \frac{x+1}{5}$. What is the y -intercept of the graph of $y = g(x)$ in the xy -plane?

(A) $(0, 1)$ (B) $(-1, 0)$ (C) $(0, 5)$ (D) $\left(0, \frac{1}{5}\right)$ **6.**

EASY · NO CALCULATOR NEEDED

A line passes through the points $(0, 2)$ and $(3, 3)$. What is the slope of the line?

7.

MEDIUM · NO CALCULATOR NEEDED

The function f is linear. Selected values of x and $f(x)$ are shown in the table. What is the value of $f(0)$?

x	$f(x)$
2	11
5	20

(A) 11

(B) 2

(C) 5

(D) 8

8.

MEDIUM · NO CALCULATOR NEEDED

The graph of $\frac{x}{4} + \frac{y}{6} = 1$ in the xy -plane is a line. What is the slope of this line?

(A) $-\frac{2}{3}$

(B) $-\frac{3}{2}$

(C) $\frac{2}{3}$

(D) $\frac{3}{2}$

9.

MEDIUM · NO CALCULATOR NEEDED

Line k is parallel to the line $y = -2x + 7$ and passes through the point $(4, 5)$. Which of the following is an equation of line k ?

(A) $y = -2x + 5$

(B) $y = \frac{1}{2}x + 3$

(C) $y = -2x + 7$

(D) $y = -2x + 13$

10.

MEDIUM · NO CALCULATOR NEEDED

Line t is defined by $y = 4x + 15$. Line s is perpendicular to line t . What is the slope of line s ?

(A) $-\frac{1}{4}$

(B) $\frac{1}{4}$

(C) 4

(D) -4

11.

MEDIUM · NO CALCULATOR NEEDED

A line passes through the points (2, 11) and (6, 27). What is an equation of the line?

(A) $y = 4x + 11$

(B) $y = 4x + 3$

(C) $y = 4x - 3$

(D) $y = 3x + 5$

12.

MEDIUM · NO CALCULATOR NEEDED

A book club charges \$13 for the first book and \$11 for each additional book. Which equation gives the total cost y , in dollars, for x books, where $x \geq 1$?

(A) $y = 11x - 2$

(B) $y = 11x + 13$

(C) $y = 11x + 2$

(D) $y = 13x + 11$

13.

MEDIUM · NO CALCULATOR NEEDED

A linear function models the total cost, in dollars, of a field trip as $C(n) = 25n + 400$, where n is the number of students attending. What does the number 25 represent in this context?

(A) Cost of each additional student

(B) Students needed to cover the fixed cost

(C) Fixed cost of the trip

(D) Cost for 25 students

14.

MEDIUM · NO CALCULATOR NEEDED

For the linear function g , $g(9) = 11$ and $g(x+1) - g(x) = 2$ for all x . Which equation defines g ?

(A) $g(x) = 2x + 9$

(B) $g(x) = 9x + 2$

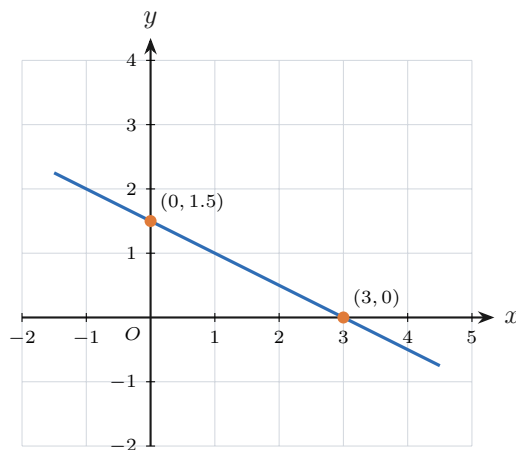
(C) $g(x) = 2x + 11$

(D) $g(x) = 2x - 7$

15.

MEDIUM · NO CALCULATOR NEEDED

The graph of a linear relationship is shown. Which equation represents the line?



(A) $y = -\frac{1}{2}x + 3$

(B) $y = 2x + \frac{3}{2}$

(C) $y = -\frac{1}{2}x + \frac{3}{2}$

(D) $y = -2x + \frac{3}{2}$

16.

HARD · NO CALCULATOR NEEDED

For the linear function h , $h(k) = c$ and $h(k + 5) = c + 15$, where c and k are constants. What is the value of $h(k + 2) - h(k)$?

17.

HARD · NO CALCULATOR NEEDED

The function g is linear, and $g(a + 3) = g(a) + 15$ for all values of a . If $g(0) = -4$, what is the value of $g(6)$?

18.

HARD · NO CALCULATOR NEEDED

If $f(x) = 4x - 3$ and $f(2a) = 3f(a) + 7$, what is the value of a ?

19.

HARD · NO CALCULATOR NEEDED

For the linear function h , $h(3) = 20$ and $h(7) = 44$. What is the value of x for which $h(x) = 62$?

20.

HARD · NO CALCULATOR NEEDED

For linear function p , $p(c) = -6$, $p(5) = 42$, and the slope of the graph of p is 8. For linear function t , $t(c) = -7$ and $t(6) = 56$. What is the slope of the graph of t ?

21.

HARD · NO CALCULATOR NEEDED

Line ℓ passes through the point $(-2, 5)$ and is perpendicular to the line $3x - 4y = 12$. What is the y -intercept of line ℓ ?

22.

HARD · NO CALCULATOR NEEDED

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the system above, t is a constant. If the system has no solution, what is the value of t ?

23.

HARD · NO CALCULATOR NEEDED

In an arithmetic sequence, the 3rd term is 17 and the 10th term is 52. What is the 20th term?

24.

SAT CHALLENGE · NO CALCULATOR NEEDED

In the xy -plane, line k has slope $\frac{2}{3}$ and passes through $(6, -1)$. Line m is perpendicular to line k and has the same y -intercept as line k . What is the x -coordinate of the x -intercept of line m ?

25.

SAT CHALLENGE · NO CALCULATOR NEEDED

For a positive constant K , the line $3x + 4y = K$ and the two coordinate axes bound a triangle of area 54. What is the value of K ?